



ANIMAL SCIENCE

Effect of mechanical processing and storage length on whole-plant sorghum silage fermentation

MARIA HELENA DE OLIVEIRA, TATIANE FERNANDES, DANIEL M. DE SOUZA, JULIANA S. BARROS, PAULO ROBERTO L. MEIRELLES & CINIRO COSTA

Abstract: Due to their smaller size, sorghum grains often remain intact during ensiling, which can impair starch availability for fermentation and reduce their overall digestibility. Thus, this study aimed to evaluate the influence of the mechanical processing of the forage mass before ensiling, as well as the length of storage in whole plant sorghum silages. The experimental design was completely randomized, in a 2×3 factorial scheme, with two processing methods (non-processed or processed) and three storage lengths (35, 70, and 105 days). Two counter-rotating smooth rolls were used to smash the chopped forage, aiming to expose the starch matrix of the sorghum grains without altering the forage fraction particle size. Crude protein contents decreased while ammonia nitrogen increased for long storage length silages due to proteolysis. *In vitro* dry matter digestibility was lower for processed silage compared to non-processed silage. Storage length increased the percentage of particles retained on sieve 3 (> 4 mm), while decreased particles retained on sieve 2 (> 9 mm) of the Penn state particle separator. The combination of processing method and storage length of whole-plant sorghum silages enhanced crude protein solubilization elucidating the effectiveness of these practices in promote starch availability.

Key words: ensiling, grain processing, nutritive value, ruminant nutrition.

INTRODUCTION

The livestock industry depends on annual crops for forage and grain feed, with corn (*Zea mays*) as the traditional crop and sorghum (*Sorghum bicolor* (L.) Moench) as an emergent, especially in scenarios of drought stress occurrence (Yang et al. 2019). In 2022, over 5.6 million tons of sorghum were used for silage production in the United States (USDA 2023), 2.5 million tons more than in the previous two years. Sorghum is a drought-tolerant crop due to different gene expressions related to water utilization (Hasan et al. 2017), making it a suitable alternative to increase water efficiency, which remains a large issue in livestock production (Roby et al. 2017).

Furthermore, sorghum has lower production costs and greater biomass yield (McCary et al. 2020), leading to great potential as a substitute for corn in animal diets without impacting animal performance (Yang et al. 2019, Lv et al. 2023).

Ample advanced technologies exist for corn silage making compared to sorghum. The use of counter-rotating rolls to reduce grain particle size is well-established for whole-plant corn silage (Ferraretto et al. 2018). This process breaks down the pericarp, exposing the grain's protein-starch matrix to fermentation conditions and enhancing digestibility (Fernandes et al. 2022). It also facilitates microbial adhesion by reducing particle size and increasing surface

area (Hoffman et al. 2011). For corn, this practice enhances starch availability for fermentation within the silo and alters the site of starch degradation within the animal's digestive system, shifting it from the small intestine to the rumen (Ferraretto et al. 2013). To our knowledge, in contrast, there is currently no farm-based equipment available to reduce sorghum grain particle size during whole-plant harvest without reducing theoretical length of cut. Then, sorghum grains are not fully broken down during the ensiling of whole-plant sorghum silage (WPSS), leading to increased amounts of whole grain which reduces starch digestibility (Yang et al. 2019). Additionally, due to their small size, sorghum grains are more likely to escape the chopping blades (Wallau et al. 2022) or chewing, remaining intact in animal feces, reducing feed efficiency.

For sorghum, long storage lengths have shown a positive effect on digestibility (Fernandes et al. 2020), resulting from increased degradation of prolamins in the protein matrix. However, there is no available data on sorghum grain processing and silage fermentation, which poses a significant challenge in terms of achieving a well-balanced diet for ruminants. Given these limitations, we utilized two counter-rotating smooth rolls to study the effect of sorghum grain breakage before ensiling. Therefore, this study aimed to evaluate the effect of grain processing and storage length on WPSS fermentation and ruminal *in vitro* dry matter digestibility (ivDMD). We hypothesized that the combination of grain processing and longer storage length would improve the fermentation and *in vitro* DM digestibility of WPSS.

Abbreviations

ADF = Acid fiber detergent
CP = Crude protein
DM = Dry matter

EE = Ether extract
ivDMD = *In vitro* dry matter digestibility
NDF = Neutral fiber detergent
NPR = Non-processed
PR = Processed
PSPS = Penn State Particle Separator
PVC = Polyvinyl chloride resin
TDN = Total digestible nutrients
WPSS = Whole-plant sorghum silage

MATERIALS AND METHODS

This article does not involve any studies with animal subjects and therefore did not require ethical approval.

Site description

The experiment was carried out at the State University of Sao Paulo, Botucatu, São Paulo, Brazil (48° 25' 28' W, 22° 51'01' S, and 777 m above sea level). The soil was a clayey, kaolinitic, thermic Typic Haplorthox (USDA 2022) with 630, 90 and 280 g/kg of clay, silt and sand, respectively. The climate is Cwa, characterized by tropical conditions with dry winters and hot, rainy summers, according to the Köppen climate classification system (Alvares et al. 2013). Monthly rainfall and temperatures during crop growth are presented in Figure 1.

Sorghum seeds (grain sorghum - 1G 233 Brevant) were sown using a no-till seed-fertilizer drill (Jumil Magnum 2580PD GII), at a depth of 0.03 m and with a row spacing of 0.5 m. At sowing 320 kg/ha of 08-28-16 N-P-K fertilizer was applied. Topdressing mineral fertilization (250 kg/ha N as urea) was performed between rows, according to the recommendations of Raij et al. (1997). Aiming to prevent the potential development of weeds, 3,000 g/ha of 6-chloro-N2 -ethyl-N4 -isopropyl-1,3,5- triazine-2,4-diamine (Atrazina Nortox 500 SC, Arapongas/ PR - Brazil) were applied 22 days after sowing. The

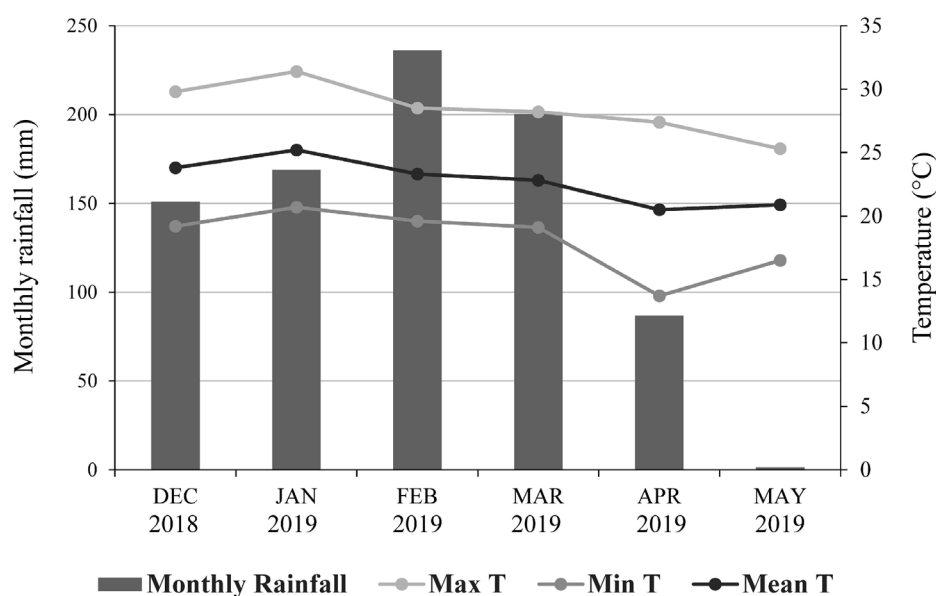


Figure 1. Monthly rainfall (mm), maximum, minimum, and mean temperatures (°C) during crop development. Source: Meteorological station FCA/UNESP – Botucatu, SP.

harvest occurred 104 days after sowing. Before harvesting all plants within 1.5 linear meters at five random points into the plot were collected for production measurements, obtaining the number of plants and their weight for green mass measurement. Subsequently, plants were manually separated into stems, leaves, and panicles for plant fractions evaluation. The dry matter (DM) was determined by drying the material in a forced circulation oven at 55°C for 72 hours. Then, the total biomass yields were calculated and extrapolated to hectares.

Silage preparation and treatments

Harvest was performed using a single-line tractor-pulled harvester, with a cutting height of 25 cm above the soil surface and chopping at 5 mm of theoretical particle size. The material was divided into 18 subsamples randomly assigned to the individual treatments and allocated in individual piles until ensilage or processing ($\pm$ 2 hours for all treatments). Treatments were composed by a 2 × 3 factorial arrangement of processing methods (non-processed – NPR; and processed - PR) and storage length (35, 70, and

105 days), with three repetitions per treatment combination. The order of ensilage followed the randomization of treatments, where NPR treatments were ensiled, and PR underwent mechanical processing before being ensiled. The fresh chopped forage was processed using two counter-rotating smooth rolls to smash the forage mass and promote the fracture and breakage of the sorghum grains without reducing the theoretical length of cut (Figure 2).

Experimental silos were made from polyvinyl chloride resin (PVC) pipes with dimensions of 400 mm in length and 100 mm in diameter. Polypropylene bags (non-toxic and semi-permeable fabric) containing 400 g of sterile sand were added to the bottom of each silo to quantify losses from effluents. Approximately 1.8 kg of forage mass was added to each experimental silo and compressed for 5 minutes using a hydraulic press, aiming to achieve a density of 600 kg/m³. Silos were weighed and sealed with PVC caps equipped with Bunsen-type valves to allow the release of gases. All experimental silos were stored at room temperature ($21.9 \pm 2.7^\circ\text{C}$) until the targeted storage length was reached.

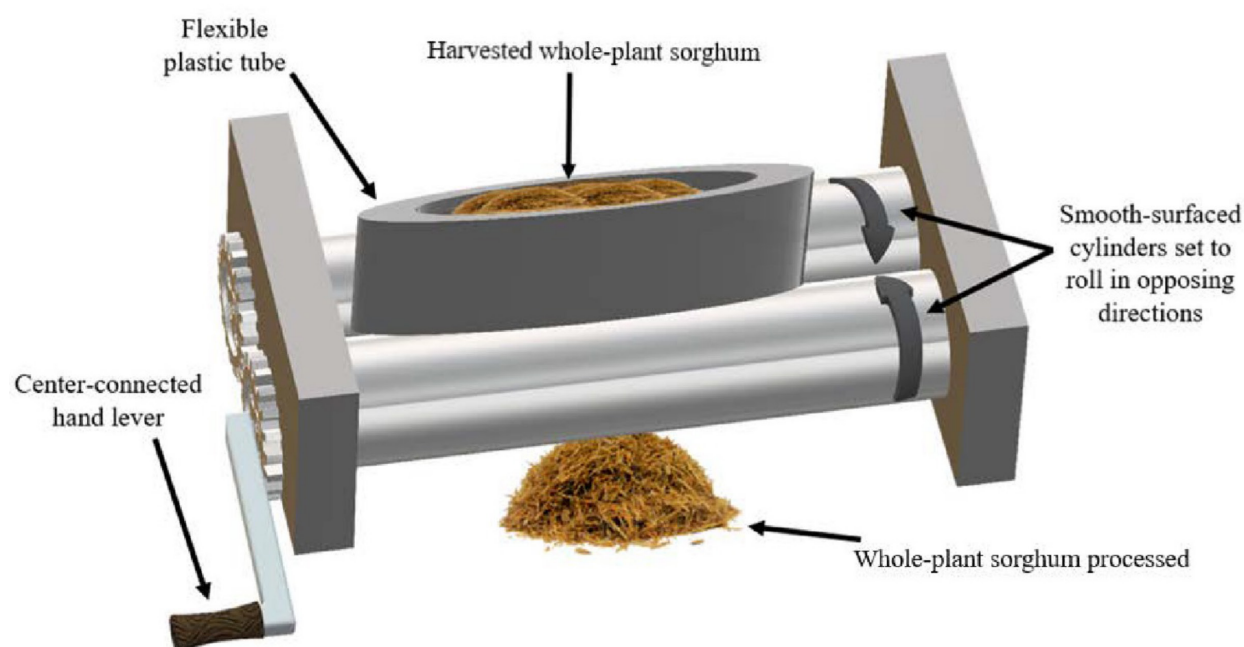


Figure 2. Counter-rotating rolls employed to fracture and break sorghum grains in the fresh chopped forage before ensiling. A flexible plastic tube was positioned at the top of the device to guide the chopped forage through the rolls during processing. The rolls were driven by a manually operated central lever.

Sample preparation and analysis

After the designated length of storage was reached, silos were weighed for the quantification of gas production, effluents, and dry matter recovery. These measurements were estimated as proposed by (Jobim et al. 2007). When opening, the upper and lower layers of the silage were discarded to avoid contamination due to possible deterioration, even though we did not observe visual signs of deterioration. The central portion of the silage was manually homogenized and divided into three subsamples for the following evaluations. The first subsample was used to determine particle size distribution using the Penn State Particle Separator (PSPS), according to Heinrichs & Kononoff (2013). The second subsample was pre-dried in a forced-air oven at 55°C for 72 hours until reaching a constant weight, then milled using a Willey-type mill equipped with a 1.0 mm sieve. The DM was determined by drying samples for 24 h at 105 °C

in a forced-air oven according to AOAC (2005), ID 976.05. Nitrogen content was determined by the Kjeldahl procedure (AOAC 2005), ID 976.05, and corrected for crude protein (CP, $N \times 6.25$). Ether extract (EE) was performed according to AOAC (2000), ID 920.39. Neutral detergent fiber (NDF) and acid detergent fiber (ADF) were determined following the methodologies of Van Soest et al. (1991). The starch content of the samples was assessed utilizing near-infrared spectroscopy FOSS NIRS DS2500TM Feed and Forage Analyzer (Foss NIR Systems, Inc., USA). Total digestible nutrients (TDN) was estimated utilizing the equation adopted by the NRC (2001), and ivDMD was determined following the ANKOM® protocol using the DAISY II incubator (ANKOM Technology Corp., Fairport, NY, USA).

The third subsample was used to assess pH. The silage extract was obtained using 30 g of fresh sample and 270 g of distilled water after homogenization in a Stomacher device (model 400 circulator, Seward Inc., Bohemia, New York,

USA) for 4 min. The measurement of pH was performed using a specific electrode coupled to a multiparameter meter (Orion Star A214 pH/ISE benchtop meter, Thermo Scientific, Waltham, MA, USA). The ammonia nitrogen content as a proportion of total nitrogen was determined by Kjeldahl distillation of the silage extract as described by AOAC (2005), ID 976.05.

Statistical analysis

Data were analyzed as a completely randomized design with 2×3 factorial arrangement of treatments, using RStudio statistical software R version 4.3.1 (R Core Team 2021). The model used is described as follows:

$$Y_{ijk} = \mu + \alpha_i + \beta_j + \alpha\beta_{ij} + \varepsilon_{ijk}$$

Where Y_{ijk} is the observed value for the response variable. μ is a constant inherent to all observations. α_i represents the effect of the i^{th} process (i = non-processed silage or processed silage). β_j represents the effect of the j^{th} storage length (j = 35, 70 or 105 days). $\alpha\beta_{ij}$ represents the

effect of the interaction between processing and storage length. ε_{ijk} is the experimental error.

We used the F-test to evaluate the effect of processing, storage length, and interaction of both. Statistical significance was declared at $P \leq 0.05$, and trend at $P \leq 0.10$. Regression analysis was performed on storage length when F-test was significant; the equation was chosen based on significance and determination coefficients (R^2).

RESULTS

A final stand of 141,000 plants/ha was obtained, with 28.430 t/ha of biomass yield and 11.325 t/ha of dry matter yield. Stalk, leaves, and panicles represented 33, 15, and 51% of DM of whole-plant sorghum, respectively.

The chemical composition of sorghum silages is presented in Table I. Processed silage had the highest ($P = 0.01$) CP content, 2.2 g/kg of DM above non-processed silage. However, both processed and non-processed silages

Table I. Composition (g/kg of DM, unless otherwise specified) of sorghum non-processed silage (NPS) or processed silage (PS) throughout days of storage length.

	Processing Method (PM)			Storage length (SL)				P value		
	NPS	PS	SEM	35	70	105	SEM	PM	SL	PM*SL
Dry matter	397.4	399.3	2.75	398.9	401.3	395.0	3.46	0.57	0.30	0.43
Crude Protein ¹	86.5	88.7	1.00	90.4	87.0	86.1	1.30	0.01	< 0.01	0.19
Ammonia g/kg of N	2.7	3.4	1.78	2.8	2.4	3.6	2.17	< 0.01	< 0.01	< 0.01
Ether Extract ²	24.7	23.7	1.68	21.1	22.0	29.5	1.74	0.52	< 0.01	0.29
Starch	372.5	373.9	2.49	375.6	373.4	370.5	4.59	0.68	0.43	0.12
Neutral Detergent Fiber	320.7	342.7	4.78	335.1	325.1	339.1	6.96	< 0.01	0.28	0.80
Acid Detergent Fiber	192.6	207.1	4.10	204.9	197.7	197.4	7.29	0.01	0.37	0.33
Total Digestible Nutrients	784.9	772.2	7.23	770.6	781.1	783.6	12.16	0.19	0.49	0.70
<i>in vitro</i> Dry Matter Digestibility	729.5	703.9	3.69	718.60	712.01	720.7	4.94	< 0.01	0.09	0.28

Significant interactions are presented in the figures. ¹Linear effect of storage length for crude protein: $Y = 91.9747 - 0.0599x$; $R^2 = 0.36$. ²Linear effect of storage length for ether extract: $Y = 15.7471 + 0.1207x$; $R^2 = 0.42$.

had a linear decrease ($P < 0.01$) throughout storage length. There was an interaction ($P = 0.01$) between processing method and storage length for ammonia nitrogen (Table I, Figure 3). Processed silage had a higher ammonia nitrogen content at 35 days and remained at the same level up to 105 days of storage. Conversely, non-processed silage had lower values at 35 days, and showed a linear increase ($R^2 = 0.71$) throughout storage length. For EE, we observed a linear increase with increasing storage length ($R^2 = 0.42$). Non-processed silage showed lower NDF ($P < 0.01$) and ADF ($P = 0.01$) content than processed silage. Consequently, the ivDMD was 25.5 g/kg lower ($P < 0.01$) for processed silage compared with non-processed silage, but there was a trend ($P = 0.09$) to increase ivDMD with increasing storage length for both silages.

The pH and DM losses of sorghum silage according to processing method and storage length are presented in Table II. There was a linear decrease with increasing storage length on pH according to storage length ($R^2 = 0.86$). An interaction between the processing method and storage length was observed for gas losses

($P < 0.01$, Figure 4). Both silages increased gas losses over storage length ($R^2 = 0.93$). However, processed silage showed a higher rate of increase in losses, resulting in greater ($P < 0.01$) losses at 105 days of storage when compared to unprocessed silage. Higher effluent loss (4.5 kg/t of GM) was observed for processed silage than non-processed silage.

The particle size distribution in the PSPS is described in Table III. There was an interaction ($P < 0.01$) between processing methods and storage length for the particle retained in the top sieve (> 19 mm) (Figure 5). Processed silage showed an increase ($P < 0.01$, $R^2 = 0.68$) in the retention at the top sieve over storage length, with the maximum retention at 105 days, with no changes ($P = 0.59$, $R^2 = 0.22$) for non-processed silage (3c). A linear decrease ($R^2 = 0.44$) with increasing storage length was observed for the particles retained in sieve 2 (> 9 mm). Proportionally, particles retained in the sieve 3 (> 4 mm) linearly increased ($R^2 = 0.39$) throughout days of storage.

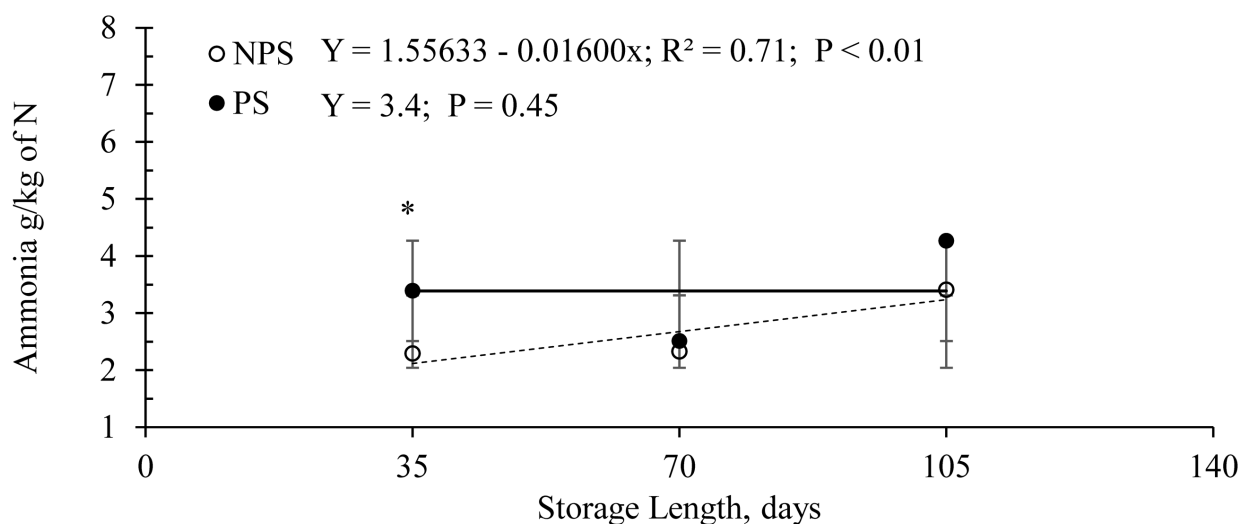
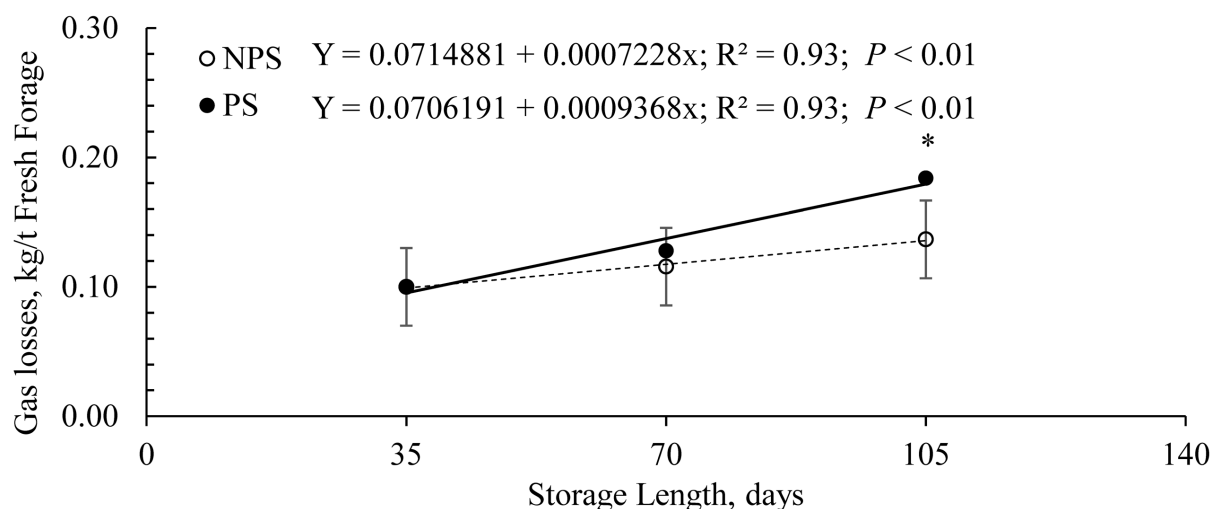


Figure 3. Interaction between non-processed silage (NPS) and processed silage (PS) with storage length for ammonia nitrogen (g/kg). * Indicates the difference between processing methods in each day of storage ($P < 0.05$).

Table II. The pH and DM losses of sorghum non-processed silage (NPS) or processed silage (PS) throughout days of storage length.

	Processing Method (PM)			Storage length (SL)				P value		
	NPS	PS	SEM	35	70	105	SEM	PM	SL	PM*SL
pH ¹	3.92	3.94	0.002	3.98	3.93	3.89	0.001	0.10	< 0.01	0.85
Gas, kg/t of fresh forage	0.12	0.13	0.014	0.10	0.13	0.16	0.007	< 0.01	< 0.01	< 0.01
Effluents, kg/t of fresh forage	17.2	21.7	1.05	19.9	19.0	19.5	1.68	< 0.01	0.85	0.77
DM recovery, %	95.5	94.6	7.26	95.1	94.9	95.2	12.12	0.39	0.97	0.10

¹Linear effect of storage length for pH: $Y = 4.026842 - 0.001323x$; $R^2 = 0.86$.

**Figure 4.** Interaction between non-processed silage (NPS) and processed silage (PS) with storage length for Gas losses (kg/t of Fresh Forage). * Indicates the difference between processing methods in each day of storage ($P < 0.05$).**Table III.** Penn State particle size distribution of sorghum non-processed silage (NPS) or processed silage (PS) throughout days of storage length.

	Processing Method (PM)			Storage length (SL)				P value		
	NPS	PS	SEM	35	70	105	SEM	PM	SL	PM*SL
Sieve 1 >19 mm	2.5	1.9	0.26	1.9	2.0	2.6	0.41	< 0.01	0.01	< 0.01
Sieve 2 ¹ > 9 mm	27.3	27.4	1.44	31.3	25.7	25.0	1.57	0.97	< 0.01	0.10
Sieve 3 ² > 4 mm	47.9	47.5	1.09	45.0	48.8	49.2	1.26	0.69	0.01	0.35
Bottom pan < 4 mm	21.7	21.8	0.72	20.5	22.1	22.6	0.75	0.61	0.07	0.06

¹Linear effect of storage length for Sieve 2: $Y = 33.5317 - 0.0883x$; $R^2 = 0.44$. ²Linear effect of storage length for Sieve 3: $Y = 43.5233 + 0.0594x$; $R^2 = 0.39$.

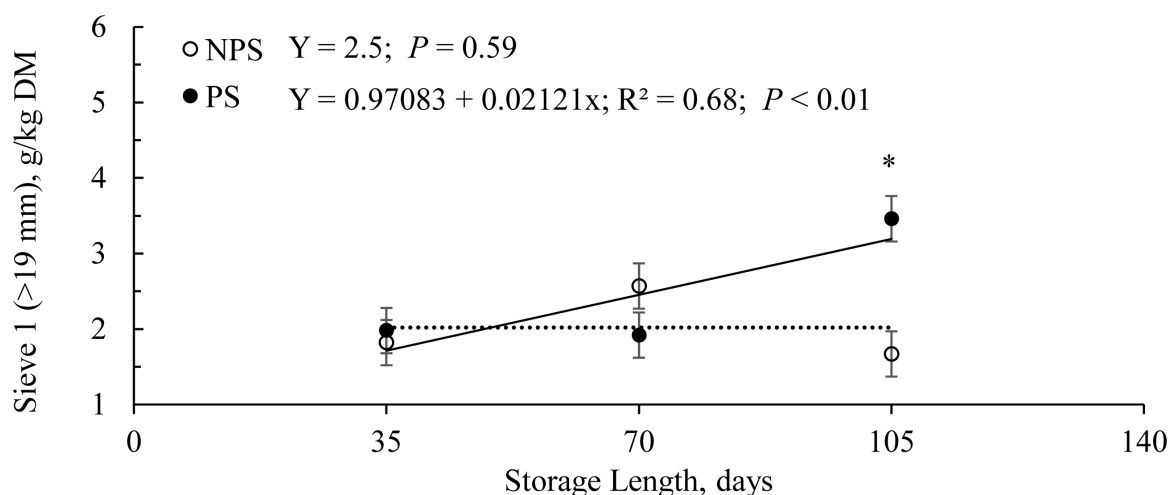


Figure 5. Interaction between non-processed silage (NPS) and processed silage (PS) with storage length for Sieve 1 (>19mm) of the PSPS (g/kg of DM). * Indicates the difference between processing methods in each day of storage ($P < 0.05$).

DISCUSSION

Grain-purpose sorghum hybrids are not often considered for silage making due to its low dry matter yield. However, this kind of hybrid was made to maximize grain yield over biomass production, resulting in a higher proportion of grains (Bean et al. 2013, Ndiaye et al. 2021), which can potentially enrich feed quality (Resende et al. 2003). In this trial, the panicles represented more than 50% of the plant on DM basis, which along with maturity at harvest increased starch deposition and played a pivotal role in increasing grain concentration over fibrous fraction. Furthermore, the high proportion of grains in the plant explains the elevated DM content, as the fibrous components may not have matured sufficiently to contribute significantly to the high DM levels observed at harvest.

Nutrient composition was affected by the increase in storage length, where the decrease in CP resulted in an increase in ammonia nitrogen. This is a natural process that occurs during ensiling, where proteins are progressively broken down into peptides and free amino acids. Subsequently, the deamination of these amino acids by microbial activity and plant

enzymes increases ammonia nitrogen (Junges et al. 2017). Our findings indicate significant protein degradation, suggesting enhanced starch availability facilitated by grain processing, which improved silage fermentation and overall quality. This mechanism has been well-documented in corn, where its prolamins, known as zeins, are solubilized in lactic acid during fermentation or broken down by bacterial proteases (Hoffman et al. 2011), resulting in improved starch digestibility (Junges et al. 2017). In grain sorghum, however, its prolamins, referred to as kafirins, are more hydrophobic (Wong et al. 2009), warranting further research to understand the extent of this process in whole-plant sorghum silage. Nonetheless, studies on rehydrated sorghum grain silage have shown that longer storage length reduced prolamins content and increased ammonia nitrogen content in a similar proportion, increasing ruminal degradability (Fernandes et al. 2022). The higher ammonia nitrogen content in the initial phase of the silage process (35 days) for processed silages, suggests that the ensiling process accelerates starch availability for degradation within a shorter storage period. This is particularly relevant at the farm level, where

faster fermentation could reduce the necessary storage time, allowing animals to access nutritionally improved forage earlier.

In this trial, mechanical processing of the silage led to greater NDF and ADF, along with reduced ivDMD. This outcome could be attributed to the compression of fibrous components, which likely caused cellular disruption and the subsequent release of cell contents during processing (Kozakai et al. 2007). The release of soluble nutrients, which may have been lost as effluent during ensiling, is supported by the higher effluent production observed in the processed silages (4.5 kg/t of fresh forage). Consequently, it contributed to an increased fiber proportion, at the expense of ivDMD. Moreover, based on the pH drop and minimal gas losses, our results indicate an absence of secondary fermentation (Borreani et al. 2018).

It is well established that corn grain particles of different sizes have different starch digestibility (Dias Junior et al. 2016), highlighting the importance of proper grain processing to enhance starch exposure for microbial attachment and fermentation in both the silo and rumen. In contrast, sorghum grain processing presents unique challenges. Sorghum grains are inherently smaller and exhibit less uniformity in size compared to corn grains, largely due to the continuous formation of sorghum panicles, which leads to grains of varying sizes and maturity stages within the same crop (Gambín & Borrás 2005). Furthermore, the mature portion of grains will present greater kafirin deposition increasing grain hardness and resistance to processing, ultimately reducing starch digestibility.

In this trial, the higher proportion of particles retained in the top sieve at 105 days of storage may be attributed to the adherence of fibrous particles or physical changes in the forage caused by compaction. These changes could potentially flatten some particles, reducing their ability

to pass through the top sieve of the PSPS, as observed by Johnson et al. (2003) and Salvati et al. (2023). Additionally, while a decrease in particles retained on sieve 2 was observed, there was a corresponding increase in particles retained on sieve 3. This is consistent with findings presented by Salvati et al. (2023), where a longer ensiling period resulted in more particles being retained in the top sieve for corn silage harvested with different forage harvesters. It was demonstrated in the same study that ensiling time from either 35 or 140 days reduced particle size of corn grains even in poorly processed silages. Herein, the absence of differences in particle accumulation in the bottom pan linked to processing method might have been related to the type of tool used, requiring an aggressive processing, with counter-rotating saw-toothed rolls for optimal efficacy, as observed in other cereal grains (Ferraretto et al. 2013).

Overall, our results suggest an increase in starch availability for degradation, based on CP and ammonia nitrogen content changes. Additionally, it showed that a smooth surface tight rolls during processing resulted in greater cellular disruption and soluble nutrient loss as effluent, which led to increased fiber proportion and reduced digestibility. These findings underscore the critical importance of addressing the unique challenges associated with sorghum processing, particularly for regions where sorghum stands out as a promising alternative forage. To fully unlock the potential of sorghum as a valuable drought-resistant and cost-effective resource for enhancing livestock nutrition, this study serves as a step forward to optimize sorghum silage production and to ensure that sorghum reaches its full potential as a reliable feed source for livestock. Moreover, this study represents a first step toward addressing the challenges associated with sorghum grain processing, underscoring the need for further research to refine processing

techniques with different roller and equipment types.

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MARIA HELENA DE OLIVEIRA^{1,2}
<https://orcid.org/0000-0003-1774-7007>

TATIANE FERNANDES³
<https://orcid.org/0000-0002-9825-5735>

DANIEL MARTINS DE SOUZA¹
<https://orcid.org/0000-0002-8328-1068>

JULIANA DA SILVA BARROS^{1,4}
<https://orcid.org/0000-0003-4612-751X>

PAULO ROBERTO DE LIMA MEIRELLES¹
<https://orcid.org/0000-0001-9168-5081>

CINIRO COSTA¹
<https://orcid.org/0000-0003-1854-2927>

¹State University of São Paulo, Department of Breeding and Animal Nutrition, School of Veterinary Medicine and Animal Science, Prof. Dr. Walter Mauricio Correa, s/n, 18618-681 Botucatu, SP, Brazil

²University of California, College of Agricultural and Environmental Science, Global Engagement Office, 1107 Environmental Horticulture Bldg, 95616, Davis, CA, U.S.A.

³School of Animal Sciences, Virginia Tech, Litton Reaves Hall, 175 West Campus Drive, 24061, Blacksburg, VA, U.S.A.

⁴State University of Piauí, Av. Dom Pedro II, 629, 64980-000 Corrente, PI, Brazil

Correspondence to: **Maria Helena de Oliveira**
E-mail: mariahelena.oliveira822@gmail.com

Author contributions

Maria Helena de Oliveira: Investigation, Methodology, Project administration, Writing – original draft. Tatiane Fernandes: Software, Validation, Writing – review & editing. Daniel Martins de Souza: Investigation, Software, Writing – review & editing. Juliana da Silva Barros: Investigation, Writing – review & editing. Paulo Roberto de Lima Meirelles: Formal Analysis, Methodology, Supervision. Ciniro Costa: Conceptualization, Funding acquisition, Investigation, Supervision, Writing – review & editing.

